

From Retrieval to Generation: Unifying External and Parametric Knowledge for Medical Question Answering

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Abstract

Medical question answering (QA) demands reliable access to rich, domain-specific knowledge. A prevalent strategy enhances large language models (LLMs) with either *external knowledge* retrieved from medical corpora or *parametric knowledge* embedded with model parameters. Existing methods generally fall into two paradigms: Retrieval-Augmented Generation (RAG), which grounds reasoning in retrieved evidence, and Generation-Augmented Generation (GAG), which relies exclusively on internally generated contexts. However, RAG often suffers from noisy or incomplete retrieval, whereas GAG is prone to hallucinated or inaccurate content due to unconstrained generation. Both limitations can misguide reasoning and compromise answer reliability. To overcome these challenges, we propose **MEDRGAG**, a unified retrieval-generation augmented framework that seamlessly integrates external and parametric knowledge for medical QA. MEDRGAG consists of two core modules: *Knowledge-Guided Context Completion* (KGCC), which steers the generator to produce background documents that complement knowledge gaps revealed by retrieval, and *Knowledge-Aware Document Selection* (KADS), which adaptively selects an optimal combination of retrieved and generated documents to construct concise yet comprehensive evidence for answers. Extensive experiments on five medical QA benchmarks demonstrate that MEDRGAG achieves a 12.5% improvement over MedRAG and a 4.5% gain over MedGENIE, highlighting the effectiveness of unifying retrieval and generation for knowledge-intensive medical reasoning. Further analyses show that our framework effectively generates complementary contexts and recovers informative retrieved evidence. Our code and data are publicly available at <https://github.com/ll0ruc/MedRGAG>.

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CCS Concepts

• Information systems → Retrieval models and ranking.

Keywords

Medical Question Answering, Retrieval-Augmented Generation, Generation-Augmented Generation

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1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across a broad spectrum of natural language understanding and question answering tasks [14, 43, 50, 52]. Nevertheless, their performance remains fundamentally constrained by the limitations of internal parametric knowledge, often resulting in hallucinations that appear coherent but are factually incorrect [16, 29]. Such factual inconsistencies pose particularly severe challenges for medical question answering (QA), a highly knowledge-intensive task that requires precise reasoning and rigorous factual correctness. In clinical scenarios, even minor hallucinations may undermine medical validity and lead to potentially harmful outcomes, highlighting the critical need for QA systems to generate reliable, evidence-grounded responses [31, 53, 55].

To improve the reliability and factual accuracy of LLM-based question answering, recent studies have increasingly focused on knowledge-augmented generation methods that ground model reasoning in external knowledge sources [8, 57]. Among these approaches, Retrieval-Augmented Generation (RAG) has emerged as a representative paradigm following a *retrieval-then-read* framework (see Figure 1 (a)). Concretely, RAG first retrieves relevant evidence from structured or unstructured medical corpora (e.g., PubMed, Wikipedia, and UMLS), and subsequently grounds the LLM's answer generation on the retrieved documents [4, 48, 56]. By incorporating up-to-date and verifiable information, RAG effectively grounds the reasoning process, thereby mitigating hallucinations and improving answer transparency [26, 30]. Despite these

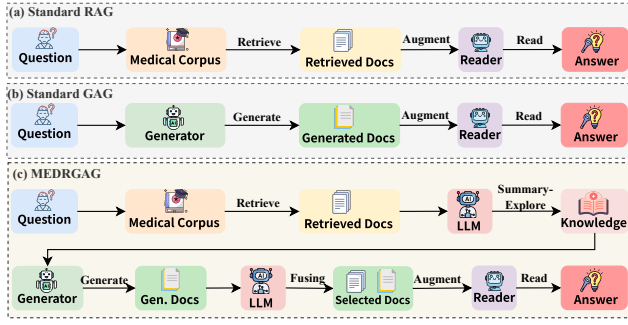


Figure 1: Comparison of standard RAG, GAG, and MEDRGAG for medical question answering.

strengths, the effectiveness of RAG is limited by two key factors: (1) retrieved documents are typically chunked into fixed-length passages that often contain noisy or irrelevant content, which can distract reasoning [4, 56]; and (2) retrieval alone may fail to provide sufficient knowledge coverage, leaving critical information gaps that impede accurate medical QA [38, 49].

In parallel with retrieval-based approaches, another line of research investigates knowledge-augmented generation by leveraging the *parametric knowledge* encoded within LLMs [5, 34]. This paradigm, known as Generation-Augmented Generation (GAG), adopts a *generate-then-read* framework (see Figure 1 (b)), in which a generator first synthesizes multiple contextual documents conditioned on the input question, and a reader subsequently utilizes these generated contexts to infer the final answer [13, 40, 51]. By removing the dependence on external corpora, this approach enables models to construct query-specific contexts that more closely align with the semantic intent of the question. Despite relying exclusively on generated documents as its knowledge source, GAG remains vulnerable to hallucinated or inaccurate content within these contexts, which can mislead reasoning and ultimately result in incorrect answers [41, 54].

Recent studies have further investigated the integration of retrieved and generated knowledge to exploit the complementary strengths of external and parametric sources [11, 54]. These methods primarily focus on mitigating knowledge conflicts that arise during the fusion of retrieved and generated documents, thereby improving question answering accuracy. Nevertheless, fully addressing the intrinsic limitations of each paradigm, including the restricted knowledge coverage of RAG and the potential overreliance on generation-based contexts in GAG, remains a challenging and unresolved problem in knowledge-intensive medical QA.

To address these challenges, we propose **MEDRGAG**, a unified retrieval-generation augmented framework that seamlessly integrates external and parametric knowledge for medical question answering. As illustrated in Figure 1 (c), MEDRGAG follows a *retrieval-generation-then-read* paradigm: it first employs a retriever to obtain relevant documents from large-scale medical corpora, then leverages a generator to produce complementary background documents, and finally adaptively fuses both sources of evidence, enabling the reader model to generate reliable answers. This design

unifies retrieval and generation within a coherent pipeline, effectively bridging external and parametric knowledge sources. Specifically, MEDRGAG consists of two core modules: (1) **Knowledge-Guided Context Completion (KGCC)**, which summarizes and analyzes retrieved documents to identify knowledge gaps required for answering the question, and subsequently guides the generator to produce targeted background documents that complement the retrieved evidence; and (2) **Knowledge-Aware Document Selection (KADS)**, which organizes retrieved and generated documents according to the question’s knowledge requirements and selects a diverse, comprehensive, and non-redundant subset of evidence for downstream reasoning. Through the coordinated operation of these two modules, MEDRGAG substantially improves knowledge completeness while reducing reliance on hallucination-prone generated contexts, thereby enhancing both factual reliability and reasoning robustness in medical QA.

We conduct extensive experiments on five widely used medical QA benchmarks, including MedQA [20], MedMCQA [32], MMLU-Med [15], PubMedQA* [21], and BioASQ [44]. We compare our approach against representative RAG-based methods (e.g., MedRAG [48]) as well as GAG-based methods (e.g., MedGENIE [13]). To further assess robustness, MEDRGAG is tested under three different reader architectures: Qwen2.5-7B-Instruct [42], LLaMA-3.1-8B-Instruct [12], and Ministral-8B-Instruct [19]. Experimental results show that MEDRGAG achieves an average accuracy gain of 12.5% over MedRAG and 4.5% over MedGENIE across all datasets and reader configurations. Further analyses and case studies demonstrate that our framework not only produces more effective complementary background documents but also successfully recovers low-similarity yet highly informative retrieved evidence.

In summary, our contributions are threefold:

- This paper propose MEDRGAG, a unified retrieval-generation augmented framework that bridges external and parametric knowledge through a coherent *retrieval-generation-then-read* paradigm for medical question answering.
- We introduce two core modules, KGCC and KADS, to complement missing knowledge and select effective evidence from retrieved and generated documents.
- The experimental results on five medical QA benchmarks show that MEDRGAG consistently outperforms a broad range of retrieval- and generation-based baselines.

2 Related Work

2.1 Medical Question Answering

Medical question answering (QA) has long been a fundamental task in biomedical natural language processing [23, 58]. Early research primarily relies on BERT-based pretrained models [10], which achieve strong performance across various medical benchmarks [1, 27]. With the rapid advancement of large language models (LLMs) such as GPT-4 [3] and LLaMA [43], their exceptional reasoning and comprehension capabilities have been increasingly extended to medical QA. To further enhance domain-specific expertise, a common strategy is to continue pretraining on large-scale biomedical corpora (e.g., HuatuoGPT [52] and PMC-LLaMA [46]). However, this approach requires substantial computational resources and high-quality annotated data, which significantly limits its practical

applicability and scalability. Consequently, knowledge-augmented QA has attracted growing attention as a more flexible and interpretable alternative [8, 57]. This paradigm incorporates external knowledge as auxiliary evidence, primarily through two representative approaches: Retrieval-Augmented Generation (RAG) and Generation-Augmented Generation (GAG).

2.2 Medical Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) enhances large language models by grounding their reasoning in external knowledge retrieved from large-scale corpora [28, 35]. MedRAG [48] establishes a comprehensive RAG benchmark and toolkit that integrates hybrid retrieval for medical corpora. Building on this foundation, *i*-MedRAG [49] introduces iterative follow-up query generation to progressively refine retrieval quality, while Self-BioRAG [18] incorporates self-reflective retrieval to better support complex multi-hop medical reasoning. More recently, Omni-RAG [7] proposes a source-planning optimization strategy to retrieve information from diverse medical resources. Despite these advances, RAG-based methods remain heavily dependent on the relevance and completeness of retrieved documents, which are often noisy and lack critical knowledge. To overcome these limitations, our framework exploits the intrinsic knowledge of large models to generate complementary background documents and adaptively recover useful retrieved evidence while filtering out irrelevant content, thereby constructing a more comprehensive and reliable evidence set for medical QA.

2.3 Medical Generation-Augmented Generation

Generation-augmented approaches prompt large language models to synthesize intermediate contexts for question answering, thereby leveraging their internal parametric knowledge [33, 36]. Representative works such as GenRead [51] and CGAP [40] show that LLMs can serve as strong context generators in open-domain QA. MedGENIE [13] extends the *generate-then-read* paradigm to the medical domain by producing multi-view artificial contexts. GRG [2] and COMBO [54] further combine generated and retrieved contexts through simple fusion strategies. Despite these advances, such methods share a fundamental limitation: the reader often over-relies on generated documents that may contain hallucinated or inaccurate information. To address this issue, our framework employs multi-step reasoning to generate complementary background documents. In addition, it integrates trustworthy retrieved evidence to improve the factuality of the final context, ultimately enabling more reliable answer generation in medical QA.

3 Methodology

3.1 Problem Formulation

Definition 3.1.1 (Medical QA). Given a medical multiple-choice question q , consisting of a question stem and an answer set $A = \{a_1, \dots, a_{|A|}\}$, the objective of the medical question answer task is to identify the most appropriate option $\hat{a}$ as the correct answer. In an LLM-based framework, the reader model $\mathcal{M}_r$ takes the question q and prompt $\mathcal{P}_r$ as input to generate an answer:

$$\hat{a} = \mathcal{M}_r(q, \mathcal{P}_r \mid \theta_r), \quad (1)$$

where θ_r denotes the parameters of the reader model.

Definition 3.1.2 (RAG). Retrieval-Augmented Generation (RAG) incorporates external knowledge into large language models to improve their reasoning capability and factual accuracy. Formally, given a medical corpus $C = \{d_1, \dots, d_{|C|}\}$ and a retriever $\mathcal{R}$ parameterized by θ_R , the retriever selects the top- k most relevant documents D_r that are most relevant to a question q :

$$D_r = \{d_1^r, \dots, d_k^r\} = \mathcal{R}(q, C \mid \theta_R). \quad (2)$$

The reader model $\mathcal{M}_r$ is further provided with the retrieved document set D_r as additional context, and generates the answer accordingly:

$$\hat{a} = \mathcal{M}_r(q, D_r, \mathcal{P}_r \mid \theta_r). \quad (3)$$

Definition 3.1.3 (GAG). In contrast, Generation-Augmented Generation (GAG) uses a large language model as a generator $\mathcal{M}_g$ to produce k tailored background documents D_g for a question q :

$$D_g = \{d_1^g, \dots, d_k^g\} = \mathcal{M}_g(q, \mathcal{P}_g \mid \theta_g), \quad (4)$$

where $\mathcal{P}_g$ denotes a generation prompt designed to encourage diversity and factual reliability, and θ_g represents the parameters of the generator. The reader model $\mathcal{M}_r$ subsequently leverages these generated documents as auxiliary context to infer the final answer:

$$\hat{a} = \mathcal{M}_r(q, D_g, \mathcal{P}_r \mid \theta_r). \quad (5)$$

3.2 MEDRGAG

We present the overall architecture of MEDRGAG in Figure 2, which unifies retrieval and generation to integrate external and parametric knowledge for medical question answering. The framework operates in three sequential stages. It first performs source-balanced evidence retrieval to collect relevant documents from multiple heterogeneous medical sources, forming an initial evidence pool grounded in external knowledge. Based on the retrieved evidence, MedRGAG then conducts knowledge-guided context completion, which identifies missing or insufficient information and generates complementary background documents to enhance knowledge coverage. Finally, the framework applies knowledge-aware document selection to jointly consider retrieved and generated documents, selecting a compact yet informative evidence set that supports accurate and reliable answer generation.

3.3 Source-Balanced Evidence Retrieval

The first stage aims to retrieve medical documents relevant to a given query from multiple heterogeneous knowledge sources. A naïve solution is to merge all sources into a single unified corpus and apply a retriever to select the top- k documents as evidence. While this strategy offers broad coverage, previous studies have shown that retrieval from an overly aggregated corpus can bias the retriever toward dominant data sources or frequently occurring information [6, 7]. Such bias ultimately compromises the fairness and diversity of the retrieved evidence.

To address this issue, we adopt a source-balanced retrieval strategy inspired by Sohn et al. [39]. Formally, given a question q and a set of source corpora $\{C_1, C_2, \dots, C_s\}$, the retriever independently selects an equal number of top-ranked documents from each source to construct an intermediate candidate set:

$$\tilde{D}_r = \bigcup_{i=1}^s \mathcal{R}(q, C_i \mid \theta_R), \quad (6)$$

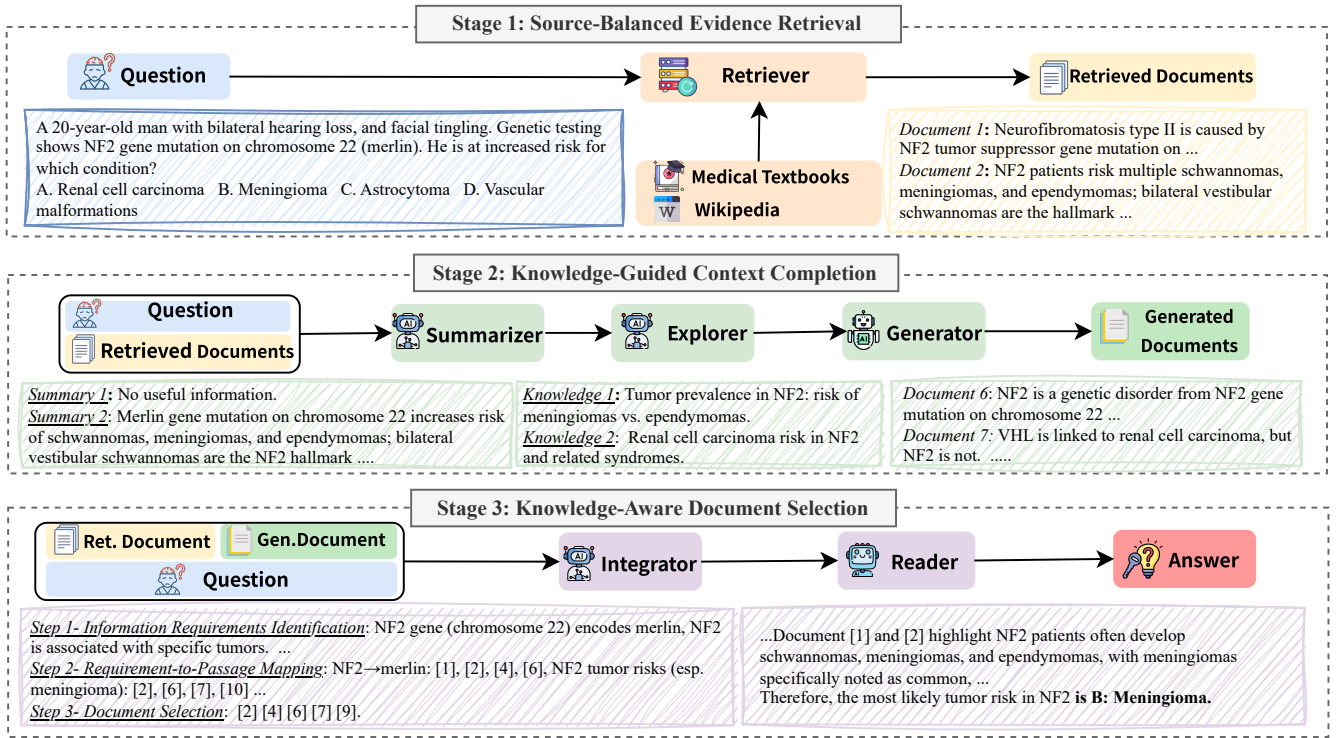


Figure 2: Framework of MEDRGAG, which comprises three stages: source-balanced evidence retrieval, knowledge-guided context completion, and knowledge-aware document selection.

which yields a more balanced and representative distribution of evidence across different corpora compared with conventional unified-retrieval approaches [28].

Following this balanced retrieval step, the candidate set $\tilde{D}_r$, containing $k \times s$ documents, is further refined via a reranking process. Specifically, we employ MedCPT-Reranker [22], an off-the-shelf cross-encoder that jointly encodes the question and each candidate document to estimate fine-grained relevance scores. The top- k documents with the highest scores in $\tilde{D}_r$ are then selected to form the final retrieved evidence set D_r for the subsequent stages.

3.4 Knowledge-Guided Context Completion

Although retrieved documents provide valuable external evidence, they suffer from two major limitations. First, the coverage of medical knowledge is often incomplete: relevant documents may capture only partial aspects of a question and therefore fail to provide comprehensive reasoning support. Second, the retrieved set frequently contains noisy or irrelevant content, which can distract the reader and ultimately degrade answer accuracy. To address these challenges, we introduce a Knowledge-Guided Context Completion (KGCC) module that generates complementary background documents, effectively targeting the insufficiency and incompleteness of retrieval-based evidence.

The KGCC process consists of three key steps: summarization of retrieved knowledge, exploration of missing knowledge, and generation of complementary background documents.

Step 1: Summarization of Retrieved Knowledge. We employ a large language model as a summarizer $\mathcal{M}_s$ to extract and condense essential information $\mathcal{D}_s$ from each retrieved document with respect to the given medical question. The summarizer produces concise knowledge summaries that retain only content relevant for answering the question, while filtering out noisy details. If a document does not contain useful information, the model explicitly outputs “No useful information.” To further enhance summarization quality, we adopt an in-context learning strategy by providing exemplar demonstrations retrieved from the training corpus to guide the model’s summarization behavior.

Summarization Prompt $\mathcal{P}_s$

You are a professional medical expert. Given the question and a retrieved document, summarize only the useful knowledge points for answering the question. If the document is irrelevant, return “No useful information”.

Step 2: Exploration of Missing Knowledge. Building on the summarized evidence, we assess its adequacy for addressing the medical question. An explorer model $\mathcal{M}_e$ is introduced to identify the most critical knowledge points that are required but absent from the current summaries, thereby ensuring that the subsequent generation stage focuses on truly complementary information. Formally, the

explorer produces a set of missing knowledge items:

$$\mathcal{K} = \{k_1, \dots, k_m\} = \mathcal{M}_e(q, D_s, \mathcal{P}_e \mid \theta_e), \quad (7)$$

where $\mathcal{K}$ denotes the set of missing knowledge points and $\mathcal{P}_e$ represents the exploration prompt. Documents labeled as “No useful information” in Step 1 are excluded to improve efficiency.

Exploration Prompt $\mathcal{P}_e$

You are a professional medical expert. Given the question and the several pieces of useful information, identify the most important missing knowledge required to answer the question thoroughly.

Step 3: Generation of Background Documents. Leveraging the identified missing knowledge points $\mathcal{K}$, the generator $\mathcal{M}_g$ produces background documents that complement the retrieved evidence. Each knowledge point $k_i \in \mathcal{K}$ serves as a guiding signal for generating a corresponding background document:

$$d_i^g = \mathcal{M}_g(q, k_i, \mathcal{P}_g \mid \theta_g), \quad (8)$$

This process yields m background documents generated from the identified missing knowledge points. If $m < k$, additional background documents are generated solely based on the question q until a total of k documents are obtained, forming $D_g = \{d_1^g, \dots, d_k^g\}$.

This design offers two key advantages. First, conditioning generation on distinct knowledge points encourages diversity among the generated documents. Second, generating supplementary documents directly from the question provides a broader contextual perspective, enriching the overall background knowledge while avoiding overfitting to localized knowledge.

3.5 Knowledge-Aware Document Selection

Given the retrieved and generated documents, a key challenge is to identify the most relevant and reliable evidence while filtering out redundant or noisy content. Prior approaches, such as GenRead [51], simply concatenate retrieved and generated documents into a single input. Although this approach increases knowledge coverage, it introduces two notable drawbacks: (1) irrelevant or low-quality documents may be retained, diluting useful information and distracting the reasoning process; and (2) the resulting longer input substantially increases the contextual burden on the reader model, leading to inefficiencies in reasoning. To overcome these limitations, we introduce the Knowledge-Aware Document Selection (KADS) module, which adaptively selects a compact yet informative subset of documents, ensuring that the reader model operates on evidence that is both reliable and comprehensive.

The proposed knowledge-aware document selection module adaptively integrates retrieved and generated evidence by aligning it with the specific knowledge requirements of each question. We employ a large language model, referred to as the integrator $\mathcal{M}_i$, to reason over all $2k$ candidate documents and select a compact yet informative subset that best supports answer generation. Concretely, $\mathcal{M}_i$ performs the following three reasoning operations:

- **Knowledge Requirement Identification.** The model analyzes the question to determine the essential knowledge components needed for a complete and accurate answer.

- **Knowledge-to-Document Mapping.** Each candidate document is mapped to one or more identified knowledge components based on its content relevance.
- **Balanced Evidence Selection.** The model then evaluates each knowledge group and selects the top- k documents that jointly maximize knowledge coverage while minimizing redundancy.

Considering the strong interdependence among the three reasoning steps, we design a carefully structured prompt that enables a single LLM to carry out the entire selection process in an integrated manner. Formally, the final evidence set D_f is obtained as:

$$D_f = \mathcal{M}_i(q, D_r \cup D_g, \mathcal{P}_i \mid \theta_i), \quad (9)$$

where $\mathcal{P}_i$ denotes the selection prompt and θ_i represents the parameters of the integrator model $\mathcal{M}_i$.

Selection Prompt $\mathcal{P}_i$

You are a medical expert. Given a question and retrieved documents, select the most useful ones for answering. Step 1. Identify key knowledge points required to answer the question. Step 2. Map each passage to these knowledge points; assign irrelevant ones to a "No Useful Information" group. Step 3. From all groups, select up to 5 passages that ensure coverage and avoid redundancy.

By dynamically balancing retrieved and generated evidence through knowledge-aware grouping, this module filters out noisy and redundant documents while retaining useful and complementary ones, leading to more efficient and accurate reasoning.

The reader model $\mathcal{M}_r$ subsequently takes the refined evidence set as input to generate the final answer:

$$\hat{a} = \mathcal{M}_r(q, D_f, \mathcal{P}_r \mid \theta_r). \quad (10)$$

To further clarify the overall workflow, we present the complete MEDRGAG framework in pseudo-code in Algorithm 1.

Algorithm 1 MEDRGAG Framework

- 1: **Input:** Question set Q , Corpus C , Retriever $\mathcal{R}$, Generator $\mathcal{M}_g$, Reader $\mathcal{M}_r$, Summarizer $\mathcal{M}_s$, Explorer $\mathcal{M}_e$, Integrator $\mathcal{M}_i$, Prompts set $\mathcal{P}$ and model parameters θ
 - 2: **Output:** Predict Answer $\hat{\mathcal{A}}$
 - 3: **for** each question $q \in Q$ **do**
 - 4: $D_r = \mathcal{R}(q, C \mid \theta_{\mathcal{R}})$ # Retrieve relevant documents
 - 5: $D_s = \mathcal{M}_s(q, D_r, \mathcal{P}_s \mid \theta_s)$ # Summarize useful information
 - 6: $\mathcal{K} = \mathcal{M}_e(q, D_s, \mathcal{P}_e \mid \theta_e)$ # Explore missing knowledge
 - 7: $D_g = \mathcal{M}_g(q, \mathcal{K}, \mathcal{P}_g \mid \theta_g)$ # Generate background documents
 - 8: $D_f = \mathcal{M}_i(q, D_r \cup D_g, \mathcal{P}_i \mid \theta_i)$ # Select documents
 - 9: $\hat{a} = \mathcal{M}_r(q, D_f, \mathcal{P}_r \mid \theta_r)$ # Produce the final answer
 - 10: **end for**
-

4 Experiments

4.1 Datasets

We evaluate MEDRGAG on five widely used and representative medical question answering benchmarks: MedQA [20], MedMCQA [32], MMLU-Med [15], PubMedQA* [21], and BioASQ-Y/N [44]. All datasets

Table 1: Main experiment results. Best results are in bold and second-best ones are underlined.

Reader	Method	MedQA-US	MedMCQA	MMLU-Med	PubMedQA*	BioASQ-Y/N	Average
Qwen2.5-7B	Direct Response	62.37	55.53	75.76	37.40	73.30	60.87
	Vanilla RAG	61.12	57.33	75.94	34.20	72.01	60.12
	MedRAG	62.61	58.64	76.12	38.00	76.38	62.35
	<i>i</i> -MedRAG	66.76	59.12	76.48	40.00	77.07	63.89
	GENREAD	68.89	60.24	78.97	46.00	79.29	66.68
	MedGENIE	69.36	59.91	78.60	<u>49.20</u>	81.88	67.79
	GRG	69.84	<u>60.89</u>	78.24	46.60	81.39	67.39
	CGAP	<u>72.35</u>	60.22	<u>79.34</u>	47.80	<u>82.36</u>	<u>68.41</u>
	MEDRGAG	75.57	62.13	81.63	51.60	84.79	71.14
Llama3.1-8B	Direct Response	67.48	58.45	75.48	55.40	76.38	66.64
	Vanilla RAG	67.24	58.38	75.85	50.20	73.30	64.99
	MedRAG	68.42	59.32	76.95	52.00	75.24	66.39
	<i>i</i> -MedRAG	70.62	60.63	77.31	53.80	76.74	67.82
	GENREAD	70.07	<u>60.89</u>	78.15	54.60	78.32	68.41
	MedGENIE	68.81	60.24	78.24	56.60	80.26	68.83
	GRG	68.97	<u>60.89</u>	76.77	57.00	82.36	69.20
	CGAP	<u>71.64</u>	60.60	<u>78.60</u>	58.20	79.94	<u>69.80</u>
	MEDRGAG	74.63	61.77	80.90	<u>57.80</u>	<u>82.04</u>	71.43
Ministral-8B	Direct Response	57.27	51.23	71.63	31.80	72.65	56.92
	Vanilla RAG	60.64	54.96	71.44	26.20	72.49	57.15
	MedRAG	62.29	56.80	76.22	29.80	74.11	59.84
	<i>i</i> -MedRAG	66.42	57.82	76.58	31.20	75.59	61.53
	GENREAD	<u>69.36</u>	59.96	77.04	41.60	78.80	65.35
	MedGENIE	68.03	59.45	<u>78.51</u>	42.00	<u>82.36</u>	<u>66.07</u>
	GRG	67.24	<u>60.08</u>	77.41	<u>42.60</u>	81.55	65.78
	CGAP	68.42	60.00	76.95	<u>42.60</u>	81.07	65.81
	MEDRGAG	74.94	62.13	80.72	44.60	84.14	69.31

Table 2: The statistics of datasets. #A.: numbers of options; Avg. L: average token counts in each question.

Dataset	Size	#A.	Avg. L	Source
MedQA-US	1,273	4	177	Examination
MedMCQA	4,183	4	26	Examination
MMLU-Med	1,089	4	63	Examination
PubMedQA*	500	3	24	Literature
BioASQ-Y/N	618	2	17	Literature

are formulated as multiple-choice QA tasks, with each question containing two to four candidate answers. Dataset statistics are summarized in Table 2, and additional details are provided in Appendix B. Following prior work [13, 48], we adopt accuracy as the primary evaluation metric for performance comparison.

4.2 Baselines

To comprehensively evaluate the effectiveness of MEDRGAG, we compare it against three representative categories of baselines. (1) **Direct response methods**, in which the LLM directly answers questions without relying on any external documents. (2) **RAG-based methods**, including Vanilla RAG [28], MedRAG [48], and

i-MedRAG [49], which retrieve relevant evidence from external medical corpora. (3) **GAG-based methods**, such as GenRead [51], MedGENIE [13], GRG [2], and CGAP [40], which synthesize auxiliary contexts from the model’s internal parametric knowledge. Detailed descriptions of these baselines are provided in Appendix C.

In our experiments, we employ BM25 [37] as the retriever to retrieve candidate documents from multiple corpora, including medical textbooks [20] and Wikipedia articles [48]. We adopt LLaMA-3.1-8B-Instruct [12] as the generator to produce background contexts. The reader is instantiated with Qwen2.5-7B-Instruct [42], LLaMA-3.1-8B-Instruct [12], and Ministral-8B-Instruct [19]. In addition, we use GPT-4o-mini [17] for the remaining LLM-based modules in MEDRGAG, including the summarizer, explorer, and integrator. For fair comparison, all retrieval, generation, and fusion stages are standardized to produce a final top-5 set of documents provided to the reader. Further implementation details are reported in Appendix A.

4.3 Main Results

The experimental results are summarized in Table 1. Overall, MEDRGAG achieves the highest average accuracy across all five medical QA benchmarks, consistently outperforming all baseline models. We highlight three key findings below.

(1) **Effectiveness over direct response methods.** MEDRGAG demonstrates substantial improvements over the direct-response

Table 3: Ablation result on Qwen2.5-7B-Instruct reader.

Method	MedQA-US	MedMCQA	MMLU-Med
Direct Response	62.37	55.53	75.76
MEDRGAG	75.57	62.13	81.63
w/o Generation	63.47	58.69	77.96
w/o Retrieval	72.90	61.15	80.35
w/o KGCC	72.27	61.30	79.43
w/o KADS	74.00	61.10	80.90

setting, where the reader model answers without external knowledge. On average, it yields an absolute accuracy gain exceeding 15%, indicating that incorporating relevant background knowledge markedly improves answer quality. Notably, this advantage is consistent across three different reader architectures, underscoring the robustness and generalizability of MEDRGAG.

(2) **Superiority to retrieval-augmented methods.** Compared with retrieval-based frameworks such as Vanilla RAG, MEDRGAG achieves an average improvement of over 9% across all reader models. Although more advanced RAG approaches, including MedRAG and *i*-MedRAG, improve retrieval quality through refined selection strategies or iterative querying, their performance gains remain relatively limited. This suggests that the medical corpora often provide insufficient knowledge coverage, as retrieved documents may fail to capture all information necessary for accurate reasoning. In contrast, MEDRGAG employs an LLM-based generator to complement knowledge gaps beyond the retrieval corpus, thereby achieving substantially higher answer accuracy.

(3) **Beyond generation-augmented methods.** MEDRGAG also outperforms generation-augmented frameworks such as MedGENIE and GenRead, achieving an average improvement of 4.5% over MedGENIE. These results indicate that integrating retrieved evidence into generation-based contexts effectively mitigates the impact of hallucinated documents and leads to more accurate reasoning. Moreover, MEDRGAG consistently surpasses GRG, which directly merges retrieved and generated documents, demonstrating that adaptive document selection enables a more precise, non-redundant, and question-relevant evidence set.

4.4 Ablation Study

We conduct ablation studies to assess the contribution of each key component in MEDRGAG. The variants are defined as follows: (1) *w/o Generation* removes the generation module and relies solely on retrieved documents for answering. (2) *w/o Retrieval* removes external retrieval and depends only on generated documents. (3) *w/o KGCC* disables the knowledge-guided context completion module, generating background documents directly without explicitly identifying missing knowledge. (4) *w/o KADS* removes the knowledge-aware document selection module and instead directly selects the top-5 documents from the ten candidates using a reranker.

As shown in Table 3, removing either the generation or retrieval component leads to a substantial performance degradation, indicating that both knowledge sources are essential for comprehensive and reliable reasoning. The performance drop is more pronounced when removing generation, suggesting that generated documents

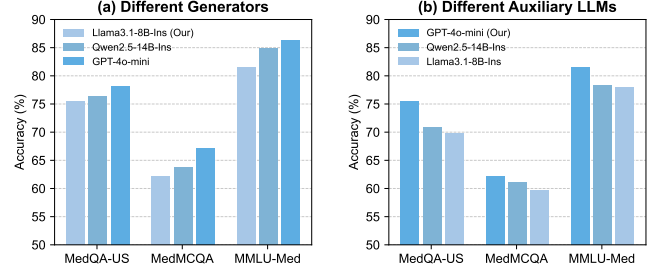


Figure 3: Results of different model scales on the Qwen2.5-7B-Instruct reader. Figures (a) and (b) show different LLM scales used as generators and summarizer-explorer-integrator.

more effectively capture question-specific knowledge than retrieved evidence alone. Disabling the KGCC module further reduces accuracy, confirming that knowledge-guided completion produces more informative and targeted contexts. Similarly, removing the KADS module also degrades performance, demonstrating that adaptive document selection plays a crucial role in identifying useful evidence for each question.

4.5 Performance Analysis

Effect of Generator Scale. We examine the impact of generator scale on overall performance by progressively increasing the parameter size from LLaMA3.1-8B-Instruct to Qwen2.5-14B-Instruct and GPT-4o-mini. As shown in Figure 3 (a), accuracy consistently improves with larger generator models, indicating that more capable generators produce higher-quality background knowledge with broader coverage. Compared with the reasoning-intensive clinical dataset MedQA-US, the performance gains are particularly pronounced on MedMCQA and MMLU-Med, which emphasize fundamental biomedical knowledge. This trend suggests that larger models encode richer underlying medical knowledge.

Effect of Auxiliary LLM Scale. We further investigate how the scale of auxiliary LLMs, which serve as the summarizer, explorer, and integrator, influences overall QA performance. In the main configuration, these auxiliary components are instantiated with GPT-4o-mini, while we additionally evaluate smaller alternatives, including Qwen2.5-14B-Instruct and LLaMA3.1-8B-Instruct. As illustrated in Figure 3 (b), performance consistently declines as the auxiliary model size decreases. The 8B and 14B models perform comparably yet remain substantially below GPT-4o-mini, indicating that smaller LLMs struggle with the complex subtasks of summarization, knowledge exploration, and document selection. Their weaker instruction-following and reasoning capabilities tend to introduce subtle yet compounding errors across stages, ultimately reducing overall answer accuracy.

Effect of Document Selection Strategies. We further analyze how different document selection strategies influence the final answer accuracy. For each question, we use five retrieved and five generated documents produced by MEDRGAG and compare several selection methods: *Random*, which randomly samples five documents from ten candidates. BGE-Reranker-Large [47] and MedCPT-Reranker [22], which rerank all candidates and select the top five documents. *Merge All*, which supplies all ten documents to the

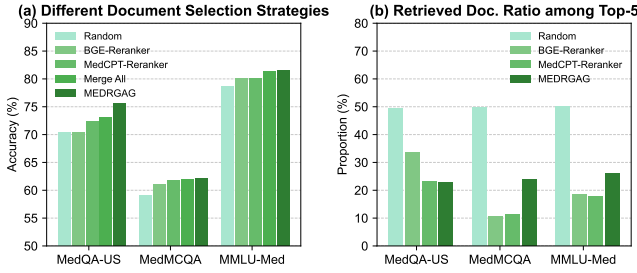


Figure 4: Results of different document selection strategies on the Qwen2.5-7B-Instruct reader. Figure (a) shows different document selection strategies on accuracy. Figure (b) shows the proportion of retrieved documents within the final top-5 across different selection strategies.

reader without filtering. As shown in Figure 4 (a), our adaptive selection strategy achieves the best overall performance across three datasets. While MedCPT slightly outperforms BGE-Reranker due to its medical-domain optimization, both remain inferior to MEDRGAG. Notably, MEDRGAG attains comparable or even higher accuracy than the *Merge All* setting while providing the reader with only half as many documents (5 instead of 10), demonstrating that adaptive document selection effectively retains useful evidence, filters out irrelevant content, and improves reasoning efficiency.

Analysis of Document Source Preference. We further examine how different document selection strategies balance the contributions of retrieved and generated evidence. For each method, we compute the proportion of retrieved documents among the final top-5 evidence, as illustrated in Figure 4 (b). Overall, generated documents tend to be favored, consistent with prior findings that ranking models often assign higher similarity scores to generative content [13, 41]. In contrast, our adaptive selection strategy substantially increases the proportion of retrieved documents, indicating that it effectively recovers valuable retrieval-based evidence that would otherwise be overshadowed by high-similarity generated passages. An exception is observed on the MedQA dataset, where our method does not increase the proportion. This is likely due to the lengthy and complex question stems in MedQA, which render retrieved passages only partially relevant rather than directly informative. Overall, these results demonstrate that MEDRGAG identifies genuinely useful evidence from both retrieved and generated sources, thereby improving QA performance.

4.6 Case Study

To further illustrate the complementary strengths of retrieved and generated knowledge in MEDRGAG, we present a representative simplified example in Table 4. For clarity, only the first two retrieved and generated documents are shown, with more detailed examples provided in Appendix E.

The question is: “A 20-year-old male with an NF2 gene mutation is at increased risk for which disease?” Among the retrieved documents, Doc 1 describes the genetic mechanism of the NF2 mutation, which is relevant but does not specify associated pathologies, while Doc 2 notes that patients with NF2 are predisposed to **meningiomas** but lacks further explanation. In contrast, the generated

Table 4: Case study showing how generated knowledge complements retrieved evidence in MEDRGAG.

Question: A 20-year-old man with bilateral hearing loss and facial tingling. Genetic testing shows NF2 gene mutation on chromosome 22 (merlin). He is at increased risk for which condition?	
Options: A. Renal cell carcinoma B. Meningioma C. Astrocytoma D. Vascular malformations.	
Retrieved Doc. 1: Neurofibromatosis type II is caused by NF2 tumor suppressor gene mutation on chromosome 22. The gene encodes merlin, which normally regulates growth factors...	
Retrieved Doc. 2: NF2 patients risk multiple meningiomas , and ependymomas; bilateral vestibular schwannomas are the hallmark. NF2 involves loss of merlin gene on chromosome 22 ...	
Generated Doc. 1: NF2 is a genetic disorder from NF2 gene mutation on chromosome 22. Patients develop bilateral vestibular schwannomas causing hearing loss, and are at increased risk for meningiomas ...	
Generated Doc. 2: VHL is linked to renal cell carcinoma , but NF2 is not. NF2 patients develop vestibular schwannomas and may also develop meningiomas and ependymomas...	
Selected Docs: [Ret_2], [Gen_1], [Gen_2]	
Final Answer: B. Meningioma ✓	

documents enrich the reasoning context: Gen 1 elaborates on the etiology of NF2 and its characteristic tumor spectrum, while Gen 2 distinguishes NF2 from **VHL**-related syndromes, thereby ruling out the distractor option “renal cell carcinoma.” Ultimately, MEDRGAG selects the three most informative documents, [Ret_2], [Gen_1], and [Gen_2], to support the reader model in producing the correct answer: “B. Meningioma”. This case demonstrates how MEDRGAG identifies missing knowledge in retrieved evidence, generates complementary documents, and selects the most relevant contexts to enable accurate reasoning.

5 Conclusion

In this paper, we propose MEDRGAG, a unified retrieval-generation augmented framework that bridges external (retrieved) and parametric (generated) knowledge for medical question answering. Unlike traditional RAG and GAG paradigms that rely on a single knowledge source, MEDRGAG integrates both through two key modules: Knowledge-Guided Context Completion (KGCC), which identifies and fills knowledge gaps exposed by retrieval, and Knowledge-Aware Document Selection (KADS), which adaptively selects the most informative evidence for reasoning. Extensive experiments on five benchmark datasets and multiple reader architectures demonstrate consistent performance improvements over strong retrieval- and generation-based baselines. Further analyses and case studies show that MEDRGAG effectively generates complementary contexts and recovers informative retrieved evidence.

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A Implementation Details

We construct the retrieval corpus from two sources: a Textbook [20] collection with 125.8K snippets and a Wikipedia [48] corpus with 29.9M snippets. We use BM25 [37] to retrieve the top-32 documents

per source, and apply MedCPT-Reranker [22] to rerank them and select the top-5 most relevant documents.

For each question, three missing knowledge points are identified. The generator, LLaMA3.1-8B-Instruct, operates in a zero-shot setting to produce up to 256 tokens per document. Two additional documents are generated directly conditioned on the question, resulting in a total of five generated contexts. The reader adopts a Chain-of-Thought (CoT) [45] prompting strategy to derive the final answer. We use GPT-4o-mini for the remaining auxiliary LLM components, including the summarizer, explorer, and integrator. The temperature is set to 1.2 for the generator and explorer to encourage diversity, and to 0.2 for the reader to maintain stability.

All baselines share identical retriever, corpus, and generator configurations for fair comparison. MedRAG [48] follows its original setup with four retrievers and Reciprocal Rank Fusion [9] for aggregation. GenRead [51] retrieves the top-1 document per question from training data and constructs five in-context learning (ICL) clusters to generate pseudo-contexts for testing. CGAP [40] generates five contexts and applies majority voting for the final prediction. GRG [2] produces ten candidate documents, from which MedCPT-Reranker selects the top-3 for fusion.

All experiments are implemented in Python 3.10 and PyTorch 2.6, running on three NVIDIA A40 GPUs (45 GB each). We employ the vLLM [25] engine for efficient batched inference and memory optimization. GPT-4o-mini is accessed via the OpenAI API for closed-source generation components.

B Dataset Details

We evaluate MEDRGAG on five representative medical QA datasets spanning clinical, biomedical, and professional knowledge domains:

- **MedQA** [20]. MedQA is derived from the United States Medical Licensing Examination (USMLE) and focuses on diagnosis, treatment, and clinical reasoning. We use the English test subset (1,273 four-choice questions).
- **MedMCQA** [32]. MedMCQA consists of Indian medical entrance-style questions covering 21 subjects and 2,400 topics. As test labels are not publicly available, the official development set (4,183 questions) is chosen.
- **MMLU-Med** [15]. MMLU-Med includes biomedical questions from six subfields, including Anatomy, College Biology, College Medicine, Clinical Knowledge, Human Genetics, and Professional Medicine, totaling 1,089 test samples.
- **PubMedQA*** [21]. PubMedQA is built from PubMed abstracts with 1,000 expert-annotated questions. To align with the RAG setting, we remove the original contexts and adopt the 500 official test samples for evaluation.
- **BioASQ-Y/N** [24, 44]. We extract all Yes/No questions from the gold-standard test sets of the machine reading comprehension (Task B) tracks over the five most recent years (2019–2023), resulting in 618 questions.

C Baseline Details

We compare MEDRGAG with a series of representative baselines from both retrieval-augmented (RAG) and generation-augmented (GAG) paradigms:

- **Direct Response.** The LLM directly answers each question without external knowledge, evaluating its intrinsic reasoning capability.
- **Vanilla RAG** [28]. A standard *retrieval-then-read* framework in which a retriever selects the top- k relevant documents from a corpus and a reader model subsequently generates the answer.
- **MedRAG** [48]. A medical RAG benchmark combining sparse and dense retrieval to obtain domain-specific evidence from MedCorp, followed by Reciprocal Rank Fusion (RRF) [9] to improve retrieval robustness and coverage.
- **i-MedRAG** [49]. An iterative extension of MedRAG that enables the LLM to issue follow-up queries, progressively refining retrieval quality and improving reasoning for multi-hop medical questions.
- **GenRead** [51]. A standard *generate-then-read* framework that first prompts an LLM to synthesize contextual documents conditioned on the input question, and then employs a reader model to generate the answer.
- **MedGENIE** [13]. The first generation-augmented framework for medical QA, which produces multi-view artificial contexts via option-focused and option-free contexts to enhance reasoning diversity and factual grounding.
- **GRG** [2]. A generator–retriever–generator pipeline where the LLM first generates hypothetical documents, retrieves supporting evidence, and then generates the final answer using both sources.
- **CGAP** [40]. A two-stage framework that performs context generation and answer prediction entirely within a single LLM, emphasizing efficiency and adaptability without relying on external corpora.

We also employ several representative large language models, retrievers, and rerankers in our experiments:

- **Qwen2** [42]. A Mixture-of-Experts model family (0.5B–72B) with strong multilingual reasoning capabilities, pretrained on large-scale corpora and instruction-tuned for general understanding.
- **LLaMA3** [12]. The latest Meta release trained on expanded data with extended context windows, achieving improved coherence and domain adaptability.
- **Mistral** [19]. An efficient open-weight model employing sliding-window attention for longer context handling with superior quality–efficiency trade-offs.
- **GPT-4** [17]. OpenAI’s flagship model trained with reinforcement learning from human feedback (RLHF), offering exceptional reasoning and instruction-following ability.
- **BM25** [37]. A classical lexical retriever using TF-IDF weighting, implemented with Pyserini to index and retrieve relevant medical snippets.
- **MedCPT-Reranker** [22]. A biomedical reranker pre-trained on 255M PubMed click logs, enabling precise semantic matching for domain-specific retrieval.
- **BGE-Reranker** [47]. A general-purpose cross-encoder reranker from BAAI that performs full query–document attention for robust semantic ranking across domains.

D Prompts Design

In this section, we present the detailed prompt templates used in MEDRGAG, covering all key components of the framework—including the summarizer, explorer, generator, integrator, and reader.

Summarization Prompt $\mathcal{P}_s$

You are a professional medical expert. Given the following question and a retrieved document, distill the useful information that can assist in answering the question. Focus only on details directly supported by evidence from the document, and avoid including irrelevant or speculative content. If the document does not contain relevant information, return “No useful information.” Do not attempt to answer the question—only summarize the essential knowledge needed for answering it accurately.

Input: Retrieved Document: {documents} Question: {question}

Output: Useful Information: {useful_information}

Exploration Prompt $\mathcal{P}_e$

You are a professional medical expert. Given the question and several pieces of useful information extracted from retrieved documents, identify the most important missing knowledge required to answer the question thoroughly. Analyze the question to determine key knowledge components, compare them with the provided information, and identify the gaps. Select the three most critical and non-redundant missing knowledge points, each expressed as a concise conceptual title rather than a full sentence.

Input: Useful Information: {information} Question: {question}

Output Format: - Reasoning: [Detailed explanation] - Knowledge 1: [Conceptual title 1] - Knowledge 2: [Conceptual title 2] - Knowledge 3: [Conceptual title 3]

Generation Prompt $\mathcal{P}_g$

You are a professional medical expert. Given the following medical question and a single knowledge point, generate a concise background document that provides relevant explanations or context strictly based on the given knowledge point. Do not infer or guess the correct answer, and avoid mentioning any answer options. Write in English and keep the content within 256 words.

Input: Question: {question} Knowledge Point: {knowledge_point}

Output: Background Document: {generated_document}

Table 5: Case study showing how generated knowledge complements retrieved evidence in MEDRGAG.**Example: #0012 in MedQA**

Question: A 20-year-old man comes to the physician because of worsening gait unsteadiness and bilateral hearing loss for 1 month. He has had intermittent tingling sensations on both cheeks over this time period. He has no history of serious medical illness and takes no medications. Audiometry shows bilateral sensorineural hearing loss. Genetic evaluation shows a mutation of a **tumor suppressor gene** on chromosome 22 that encodes merlin. This patient is at increased risk for which of the following conditions?"

Options: A. Renal cell carcinoma B. Meningioma C. Astrocytoma D. Vascular malformations.

Retrieved Doc. 1: Neurofibromatosis. Neurofibromatosis type II is caused by a **mutation on chromosome 22**. The mutation falls on the NF2 tumor suppressor gene. The gene normally encodes a cytoplasmic protein known as merlin... The loss of function leads to increased activity of growth factors normally regulated by merlin, leading to the formation of the tumors associated with NF2.

Retrieved Doc. 2: Pathology_Robbins. Most schwannomas are sporadic, but about 10% are associated with familial **neurofibromatosis type 2** (NF2). NF2 patients are at risk of developing multiple schwannomas, **meningiomas, and ependymomas** (the latter are described in Chapter 23). The presence of bilateral vestibular schwannomas is a hallmark of NF2; despite the name, neurofibromas (described later) are not found in NF2 patients...

Summary Doc. 1: No useful information.

Summary Doc. 2: Patients with a mutation of the merlin gene on chromosome 22 are at risk of developing multiple schwannomas, meningiomas, and ependymomas. The presence of bilateral vestibular schwannomas is a hallmark of neurofibromatosis type 2 (NF2).

Explore Knowledge 1: Specific tumor prevalence in NF2 patients (e.g., risk of meningiomas vs. ependymomas).

Explore Knowledge 2: Risk of renal cell carcinoma in NF2 and related syndromes.

Generated Doc. 6: ****Neurofibromatosis Type 2 (NF2) **** **Neurofibromatosis Type 2** (NF2) is a rare genetic disorder characterized by the development of non-cancerous (benign) tumors in the nervous system. The condition is caused by mutations in the NF2 gene, also known as the **merlin tumor suppressor gene**, which is located on chromosome 22. ****Clinical Features:**** NF2 patients often present with bilateral vestibular schwannomas (tumors of the nerves responsible for hearing and balance) leading to sensorineural hearing loss...

Generated Doc. 7: ****Von Hippel-Lindau Syndrome and Neurofibromatosis Type 2 (NF2) Association with Renal Cell Carcinoma **** Von Hippel-Lindau syndrome (VHL) is a genetic disorder characterized by the development of benign and malignant tumors, particularly hemangioblastomas of the cerebellum, retina, and spinal cord. The NF2 gene encodes for merlin (also known as schwannomin), a tumor suppressor protein that plays a critical role in the regulation of cellular growth and division...

Selected Docs: Step 1: Information Requirements Identification: NF2 gene (chromosome 22) encodes merlin, NF2 is associated with specific tumors. Step 2: Requirement-to-Passage Mapping: NF2→merlin: [1], [2], [4], [6], [9]; NF2 tumor risks (esp. meningioma): [2], [6], [7], [9], [10]; Other tumors (e.g., ependymoma/astrocytoma): [2], [5], [4], [6], [7] Step 3: Document Selection for Completeness and Conciseness: Prioritize NF2→meningioma evidence; ensure RCC exclusion. Final Selection: [2] [4] [6] [7] [9].

Final Answer: B. Meningioma ✓

E More Case**Selection Prompt $\mathcal{P}_i$**

You are a professional medical expert. Given a medical question and ten candidate passages (each labeled with an identifier [id]), select the top-5 most useful passages for answering the question accurately.

Follow the reasoning steps below: 1. **Information Requirements Identification:** Identify the key knowledge points necessary to answer the question thoroughly. 2. **Requirement-to-Passage Mapping:** Match each passage to the corresponding knowledge point(s) and classify irrelevant ones into a "No Useful Information" group. 3. **Document Selection for Completeness and Conciseness:** Choose up to five passages that together provide comprehensive coverage of the key knowledge points while minimizing redundancy.

Input: Documents: {documents} Question: {question}

Output Format: - Reasoning: [Detailed explanation] - Final Selection: [id1] [id2] [id3] [id4] [id5]

Answering Prompt $\mathcal{P}_r$

You are a professional medical expert. Given the question and several retrieved or generated documents, reason step-by-step and provide the final answer. First, extract and utilize useful information from the documents; if insufficient, rely on your medical knowledge to complete the reasoning. Return your output in JSON format containing both reasoning and the final answer choice.

Input: Retrieved Documents: {documents} Question: {question}

Output Format: {"reasoning": "explanation", "answer_choice": "A/B/C/..."}